ALGEBRAICITY GOES UP IN TOWERS OF FIELD EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Cited by the algebraic-closures-isomorphic lemma (b7 card L48373) to see that the image of a closure sits algebraically inside the target closure.

Editorial note. The label lemma-algebraic-goes-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Definition of an algebraic element: alpha is algebraic over a base when some nonzero polynomial with coefficients in the base vanishes at alpha
- For a tower F inside E inside K, polynomials over F are polynomials over E

Lemma 0.1. *Let $K/E/F$ be a tower of field extensions.*

(1) *If $\alpha \in K$ is algebraic over F, then α is algebraic over E.*
(2) *If K is algebraic over F, then K is algebraic over E.*

Proof. This is immediate from the definitions. □

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